Why Domain Matters, part 2

Logistics and supply chain

The record says

`Delivered on time, in full.`

What actually happened

One pallet line held the wrong pack size.

Monday 22 June, 6:12 am. Store 014, Jayanagar.

The truck made its slot. Every record says the delivery was in full.

```
Goods receipt GRN-260622, 06:12
Pallet …0542 scanned, notice matched
Line 1: atta 5 kg, 16 cases, 64 bags
Discrepancies: none
```

On line 1, the 16 bales hold 1 kg packs, 160 of them. The 5 kg bags are still in the warehouse.

Five readings of one delivery

Data product	What it reads	What it concludes
OTIF dashboard	88% for June; misses on 11, 13, 15 June, none of them atta	“22 June was perfect.”
Data tests	10 of 10 pass: notice, picks, pallets and receipts join without a mismatch	“The pipeline is green.”
Inventory reports	Four atta corrections in two buildings, netting to ₹80	“Nothing to chase.”
Store loss report	26 June count: 5 kg −63, 1 kg +157	“Sloppy counting, or theft.”
AI assistant, given the schema	On time, inside the slot, every line booked as noticed	“On time and in full.”

Each conclusion is reasonable given the records in front of it. Each misses the wrong line.

The scan confirmed the slot, not the item

Pick confirmation	`03:00, slot A-07-03: 16 cases of atta 5 kg, by slot label scan`
Shipping notice	`Line 1: 16 cases of atta 5 kg, 64 bags`
Pallet label	`SSCC 029000010000010542`
Goods receipt	`GRN-260622, 06:12: 64 bags of atta 5 kg booked`
On the pallet	16 bales of atta 1 kg, 160 packs

At 03:00 the picker stood at slot A-07-03 (5 kg) and took the bales from A-07-04 next to it. Every record after that was written from the pick confirmation, so they all agree.

Nothing in the records says “wrong item”

Pallet	Line	Item	Cases	Units
…0542	1	Atta 5 kg	16	64
…0542	2	Atta 1 kg	9	90
…0542	3	Rice 5 kg	14	56
…0559	4	Sugar 1 kg	5	50

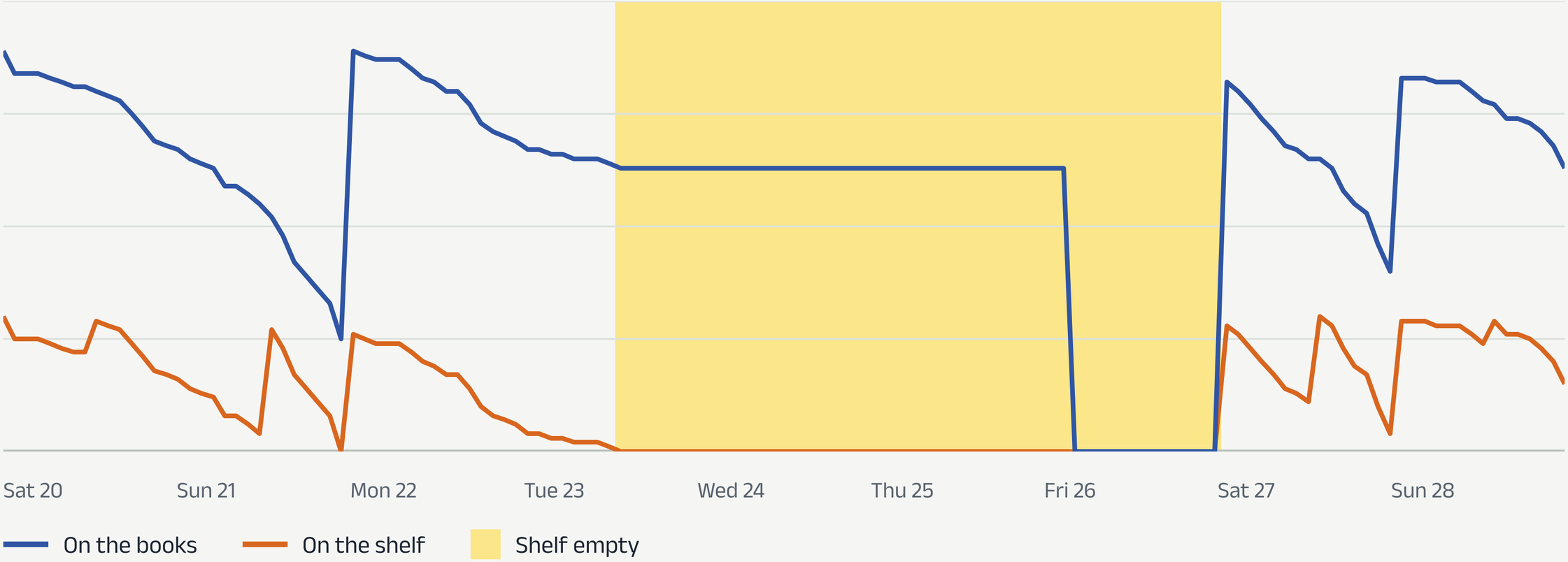
Receipt: booked as notice. Discrepancies: none.

The notice, the pallet label and the receipt join without a mismatch, because all three come from the same pick.

The evidence is in records that do not share that source: pallet weights against case weights, a 1 kg backroom growing without receipts, 5 kg sales stopping while the books sit at 63.

Shipping notice SHP-260622 and goods receipt GRN-260622. Identifiers use GS1 prefix 29, set aside for restricted circulation.

The books said 63. The shelf was empty.



5 kg atta at Store 014, bags, every open hour from 20 to 28 June 2026. Grid lines every 25 bags. Synthetic world, seed 42.

What the shelf lost

51.5

open hours with the 5 kg shelf empty

₹19,080

of 5 kg atta customers wanted and could not buy

72 bags wanted during the gap and not bought

What the shopper did	Bags
Went to another store	29
Took a different pack size	16
Bought another brand	12
Came back later	9
Never bought it	6

At the shelf price of ₹265 a bag. Shopper responses follow Gruen, Corsten and Bharadwaj (2002).

Four corrections, one mistake

Where and when	Item	Units	At cost	Recorded as
Warehouse, Tue 23 Jun 14:20	Atta 1 kg	−160 (−16 cases)	−₹7,920	slot count variance
Warehouse, Thu 25 Jun 10:00	Atta 5 kg	+64 (+16 cases)	₹14,592	cycle count variance
Store, Fri 26 Jun 09:00	Atta 5 kg	−63	−₹14,364	count after complaints
Store, Fri 26 Jun 09:00	Atta 1 kg	+157	₹7,772	count after complaints
Net, across the company			₹80	

Equal and opposite variances in neighbouring slots, mirrored at a store that got a truck from them, is the signature of a mis-pick.

The pipeline is green

10 of 10

data tests pass

Keys, joins and reconciliations on the shipping records. They agree with each other, because every one was written from the same pick confirmation.

0 of 5

tests about how a warehouse works pass

- No correction has a mirror image in the same building
- No warehouse correction is mirrored at the store
- Every receipt survives the next count
- Something other than the pick checks each pallet
- No silence longer than the item’s usual rate allows, while the books show stock

All 15 tests use only the records. The one about pallets fails for all 30 of June’s: none was weighed or case-scanned.

Pair corrections before you explain them

₹80

the four atta corrections, netted. Gross: ₹44,648.

5 pairs

equal and opposite in cases: 4 atta, in each building and across the two; 1 milk, the till’s scan errors

The trace: 16-case picks of 5 kg from slot A-07-03 in the week before the first variance

```
2026-06-22 03:00 SHP-260622 line 1 SSCC 029000010000010542 16 cases slot label scan
```

One row. The answer key: that line held 160 packs of 1 kg.

Mirror pairs: one product family, opposite signs, each at least a case, within 15% in cases, within a week. The warehouse’s pair was complete 3 days after the truck, a day before the store’s count.

Give the assistant a domain layer

1 → 7 of 7

questions answered right at the supplier review, schema only → with the domain layer

- “An SSCC identifies a pallet, not what is on it”
- “A goods receipt booked by label scan is a copy of the notice”
- Pick confirmations, pallets, warehouse corrections, till lines
- Five views, the mirror pairs among them

Was the 22 June delivery on time and in full? With the domain layer:

On time, at 06:12, but not in full. Line 1, 16 cases on the notice as 5 kg atta, held 1 kg packs: the warehouse's 1 kg slot came up 16 cases short on Tue 23 Jun and its 5 kg slot 16 cases over on Thu 25 Jun, and the store's count on Fri 26 Jun found 63 bags of 5 kg missing and 157 more 1 kg packs than the books showed. It was the only 16-case 5 kg pick in the week before.

The assistant is scripted so the run can be repeated; the technical companion shows every query.

This is the normal case

99.68%

best-in-class order-picking accuracy (WERC DC Measures, 2025). About one in 300 still goes wrong.

57%

of supplier orders physically checked at retail warehouses, on average; about 1 retailer in 10 checks every unit

92%

median accuracy those checks found, against a median goal of 98%

Receiving by scanning a pallet's SSCC against the shipping notice is how GS1's standards are meant to work. It is exactly as accurate as the notice.

Sources: WERC DC Measures 2025 via Yale; Gibson (2017), Auburn University ASN study; GS1 UK, Logistics labels and SSCCs.

One name, four numbers

92%

The DC's report

Dispatched on schedule, shipped as ordered, by its own pick confirmations

90%

Scorecard, per line

Order lines in the slot and in full, by the store's goods receipt

88%

Scorecard, per delivery

Deliveries in the slot with every line in full, by goods receipt

84%

What the shelf got

Deliveries in the slot with every case what was ordered

OTIF for the warehouse's 25 deliveries to Store 014 in June. The 22 June delivery passes the first three and fails the fourth. Reconciled with the counts, the records give 84% too, after the fact.

Two of three fixes make OTIF worse

Scan every case at pick

88% → 88%

scorecard unchanged, and now true: the shelf got 88% too; 1 bag missed all week

The scanner reads the bale, not the slot, and rejects 1 kg on a 5 kg line.

Weigh every pallet

88% → 84%

scorecard falls; 1 bag missed all week

The pallet is 160 kg light. Re-picked, the truck arrives 06:42, late.

Scan every case at the store

88% → 84%

scorecard falls; 0 bags missed

The clerk books what arrived and raises a claim. 5 kg is reordered that night.

OTIF per delivery for June, on the scorecard. Same customers and trucks in every scenario; the changes start on 1 June. Try them in the demo.

Five questions to ask your team

1. Which of our records were written from the same event, and which observe the goods independently?
2. Do we net our stock corrections, or pair them?
3. When a KPI says “in full”, which record says so, and could anything have contradicted it?
4. Can our data trace a variance back to the order line, the pallet and the pick that caused it?
5. What does our AI assistant know about how a warehouse works, and how do we test that?

For data and AI leaders. The technical companion answers each one for the 22 June truck.

Every record agreed. That is why nobody checked the pallet.

The tests in SQL, the pairs, the trace and a notebook on the records:

`kvinayakpai.github.io/domain-series/logistics/technical`

> **Next: payments**
>
> The till says the payment failed. The customer’s bank says it went through.